

JARVIS

A modular, voice-first personal AI agent. Phone-first, cloud/free services only, \$0 baseline.

Current milestone: VO (Speak → AI → spoken answer)

Tap the big button, talk, JARVIS answers out loud. Nothing else yet (no wake word, no tools, no long-term memory).

How it works

Tap → microphone → speech-to-text →
Brain → AI provider → text → text-to-
speech → speaker
 (voice.js)
(brain.js) (providers.js) (voice.js)

File	Layer	Job
<code>index.html</code> , <code>style.css</code> , <code>app.js</code>	Interface	Page, button, settings screen

File	Layer	Job
<code>controller.js</code>	Interface logic	One tap = listen → think → speak; cancel/stop
<code>voice.js</code>	Voice	Phone's built-in speech recognition + speech synthesis
<code>brain.js</code>	Brain	Conversation (short-term memory), system prompt
<code>providers.js</code>	Provider	Adapters: OpenAI-compatible (Groq, OpenAI, OpenRouter...) and Gemini
<code>config.js</code> , <code>errors.js</code>	Support	Settings in the phone's browser; friendly error messages
<code>jarvis.test.mjs</code>	Tests	Automated tests with fake network/mic/speaker

Adding another AI provider later (e.g. Claude) means adding one small class in `providers.js`

and one entry in `config.js`. Nothing else changes.

Setup (phone only)

1. Free API key. Easiest: Groq. Open <https://console.groq.com/keys>, sign up (no card), tap *Create API Key*, copy it.
2. GitHub. Create a free account, then a Public repository named `JARVIS` (GitHub Pages is free only for public repos). Upload all files from this folder (*Add file* → *Upload files*). Create a file named `.gitignore` and paste the contents of the `.gitignore` file.
3. Publish. Repository → *Settings* → *Pages* → Source: *Deploy from a branch* → Branch `main`, folder `/ (root)` → Save. After 1–2 minutes the page is at `https://YOUR-USERNAME.github.io/JARVIS/`.
4. Open that address in Chrome (Android) or Safari (iPhone), open *Settings* inside JARVIS, paste your key, Save.
5. Tap the button, allow the microphone, say "Hello JARVIS".

Configuration

There are no environment variables: the key is typed into the app and stored only in your phone's browser storage. It is never in the repository. `.gitignore` also blocks `.env`, key files and credentials in case you add any later.

Free-tier facts (checked 2026-10-05)

- Groq free plan: no card; limits per model, e.g. `openai/gpt-oss-20b` 30 requests/min, 1,000/day. Check your own limits at console.groq.com/settings/limits.
- Gemini free tier: no card, but Google no longer publishes fixed numbers (see your AI Studio rate-limit page), and free-tier requests may be used to improve Google products.
- Speech: done by the browser. On Chrome for Android the audio is processed by Google's speech service.
- GitHub Pages: free for public repositories, HTTPS included (needed for the microphone).

Tests

`node --test` (needs Node.js 18+; optional, only on a computer or a free GitHub Action). Covers

providers, errors, brain, speech helpers, controller flow, settings and a scan for hard-coded keys.

Current limitations

- Real microphone/speaker behaviour depends on your phone's browser and can only be confirmed on the phone.
- One question per tap; conversation context is remembered until you reload the page or tap *New conversation*.
- iPhone: use Safari in a normal tab; speech recognition is less reliable on iOS than on Android Chrome.
- Model names change over time. If you see "model name may be wrong", edit the *Model* field in Settings.

Roadmap

V1 continuous conversation + interruption → V1.5
wake word → V2 memory → V3 tools (email,
calendar, files, web) → V4 planner → V5
specialist agents → V6 browser/mobile control

→ V7 safety (permissions, confirmations, logs, emergency stop).